

Regularization - 15 Numerical Solutions

Part A: Regularization - 10 Questions

Q1. L1 Regularization

Given: $w_1 = 2$, $w_2 = -3$, $\lambda = 0.5$, Original Loss = 10.

Step 1: Calculate the L1 penalty.

$$\begin{aligned} |w_1| + |w_2| &= |2| + |-3| \\ &= 2 + 3 \\ &= 5 \end{aligned}$$

Step 2: Calculate the regularization penalty.

$$\begin{aligned} \lambda \times 5 &= 0.5 \times 5 \\ &= 2.5 \end{aligned}$$

Step 3: Calculate total loss.

$$L_{\text{total}} = 10 + 2.5$$

Final Answer: $L_{\text{total}} = 12.5$

Q2. L2 Regularization

Given: $w_1 = 2$, $w_2 = -3$, $\lambda = 0.1$, Original Loss = 5.

Step 1: Calculate $\sum w_i^2$.

$$\begin{aligned} 2^2 + (-3)^2 &= 4 + 9 \\ &= 13 \end{aligned}$$

Step 2: Calculate the regularization penalty.

$$0.1 \times 13 = 1.3$$

Step 3: Calculate total loss.

$$L_{\text{total}} = 5 + 1.3$$

Final Answer: $L_{\text{total}} = 6.3$

Q3. L1 vs L2

Given: $w_1 = 3$, $w_2 = -4$, $w_3 = 2$, $\lambda = 0.2$, Original Loss = 8.

Step 1: L1 penalty.

$$\begin{aligned} 0.2 \times (|3| + |-4| + |2|) \\ &= 0.2 \times (3 + 4 + 2) \\ &= 0.2 \times 9 \end{aligned}$$

L1 penalty = 1.8

Step 2: L2 penalty.

$$\begin{aligned} 0.2 \times (3^2 + (-4)^2 + 2^2) \\ &= 0.2 \times (9 + 16 + 4) \\ &= 0.2 \times 29 \end{aligned}$$

L2 penalty = 5.8

Step 3: L1 regularized loss = 8 + 1.8

L1 regularized loss = 9.8

Step 4: L2 regularized loss = 8 + 5.8

L2 regularized loss = 13.8

Q4. Ridge Regression

Given: $w_1 = 4$, $w_2 = 3$, $MSE = 6$, $\lambda = 0.5$.

Step 1: Calculate $\sum w_i^2$.

$$4^2 + 3^2 = 16 + 9$$

$$= 25$$

Step 2: Calculate penalty.

$$0.5 \times 25 = 12.5$$

Step 3: Calculate Ridge loss.

$$L_{\text{Ridge}} = 6 + 12.5$$

Final Answer: $L_{\text{Ridge}} = 18.5$

Q5. Lasso Regression

Given: $w_1 = 5$, $w_2 = -2$, $w_3 = 1$, $\text{Loss} = 8$, $\lambda = 0.4$.

Step 1: Calculate $\sum |w_i|$.

$$|5| + |-2| + |1| = 5 + 2 + 1$$

$$= 8$$

Step 2: Calculate penalty.

$$0.4 \times 8 = 3.2$$

Step 3: Calculate total loss.

$$L_{\text{Lasso}} = 8 + 3.2$$

Final Answer: $L_{\text{Lasso}} = 11.2$

Q6. L2 Weight Update

Given: $w = 5$, $\eta = 0.1$, $\partial L / \partial w = 2$, $\lambda = 0.5$.

Step 1: Use $w_{\text{new}} = w - \eta(\partial L / \partial w + 2\lambda w)$.

$$2\lambda w = 2 \times 0.5 \times 5$$

$$= 5$$

Step 2: Calculate the total gradient.

$$2 + 5 = 7$$

Step 3: Update the weight.

$$w_{\text{new}} = 5 - 0.1 \times 7$$

$$= 5 - 0.7$$

Final Answer: $w_{\text{new}} = 4.3$

Q7. L1 Weight Update

Given: $w = 4$, $\eta = 0.1$, $\partial L / \partial w = 3$, $\lambda = 0.5$.

Since w is positive, use $w_{\text{new}} = w - \eta(\partial L / \partial w + \lambda)$.

$$w_{\text{new}} = 4 - 0.1(3 + 0.5)$$

$$= 4 - 0.1(3.5)$$

$$= 4 - 0.35$$

Final Answer: $w_{\text{new}} = 3.65$

Q8. Ridge with Multiple Weights

Given: $w_1 = 2$, $w_2 = -5$, $w_3 = 3$, $\text{MSE} = 12$, $\lambda = 0.2$.

Step 1: Calculate $\sum w_i^2$.

$$2^2 + (-5)^2 + 3^2 = 4 + 25 + 9$$

$$= 38$$

Step 2: Calculate penalty.

$$0.2 \times 38 = 7.6$$

Step 3: Calculate total loss.

$$L_{\text{Ridge}} = 12 + 7.6$$

Final Answer: $L_{\text{Ridge}} = 19.6$

Q9. Lasso with Multiple Weights

Given: $w_1 = -4$, $w_2 = 6$, $w_3 = -2$, $\text{MSE} = 10$, $\lambda = 0.3$.

Step 1: Calculate $\sum |w_i|$.

$$|-4| + |6| + |-2| = 4 + 6 + 2$$

$$= 12$$

Step 2: Calculate penalty.

$$0.3 \times 12 = 3.6$$

Step 3: Calculate total loss.

$$L_{\text{Lasso}} = 10 + 3.6$$

Final Answer: $L_{\text{Lasso}} = 13.6$

Q10. Elastic Net

Given: $w_1 = 2$, $w_2 = -3$, $\text{Loss} = 7$, $\lambda = 0.5$, $\alpha = 0.6$.

Step 1: Calculate $\sum |w_i|$.

$$|2| + |-3| = 5$$

Step 2: Calculate $\sum w_i^2$.

$$2^2 + (-3)^2 = 4 + 9 = 13$$

Step 3: Calculate penalty.

$$0.5[0.6(5) + 0.4(13)]$$

$$= 0.5[3 + 5.2]$$

$$= 0.5 \times 8.2$$

$$= 4.1$$

Step 4: Calculate total loss.

$$L_{\text{total}} = 7 + 4.1$$

Final Answer: $L_{\text{total}} = 11.1$

Part B: Cost Function + Regularization - 5 Questions

Q1. L2 Regularized Cost Function

Given: $m = 5$, errors = 2, -1, 3, -2, 1; $\theta_1 = 2$, $\theta_2 = -3$; $\lambda = 0.4$.

Step 1: Calculate the sum of squared errors.

$$2^2 + (-1)^2 + 3^2 + (-2)^2 + 1^2$$

$$= 4 + 1 + 9 + 4 + 1$$

$$= 19$$

Step 2: Calculate original cost.

$$J(\theta) = 19/(2 \times 5)$$

$$= 19/10$$

Original cost = 1.9

Step 3: Calculate $\sum \theta_j^2$.

$$2^2 + (-3)^2 = 4 + 9 = 13$$

Step 4: Calculate L2 penalty.

$$\lambda/(2m) \times 13 = 0.4/10 \times 13$$

$$= 0.52$$

Step 5: Calculate total cost.

$$J_{\text{total}}(\theta) = 1.9 + 0.52$$

Final Answer: $J_{\text{total}}(\theta) = 2.42$

Q2. L1 Regularized Cost Function

Given: $m = 4$, errors = 1, -2, 2, -1; $\theta_1 = 3$, $\theta_2 = -4$; $\lambda = 0.5$.

Step 1: Sum of squared errors.

$$1^2 + (-2)^2 + 2^2 + (-1)^2 = 1 + 4 + 4 + 1$$

$$= 10$$

Step 2: Original cost.

$$J(\theta) = 10/(2 \times 4) = 10/8$$

Original cost = 1.25

Step 3: Calculate $\sum |\theta_j|$.

$$|3| + |-4| = 7$$

Step 4: Calculate L1 penalty.

$$0.5/(2 \times 4) \times 7 = 0.5/8 \times 7$$

$$= 0.4375$$

Step 5: Total cost.

$$J_{\text{total}}(\theta) = 1.25 + 0.4375$$

Final Answer: $J_{\text{total}}(\theta) = 1.6875$

Q3. Compare Regularized and Unregularized Cost

Given: $m = 5$, squared errors = 4, 1, 9, 4, 1; $\theta_1 = 2$, $\theta_2 = -2$, $\theta_3 = 1$; $\lambda = 0.5$.

Step 1: Sum of squared errors.

$$4 + 1 + 9 + 4 + 1 = 19$$

Step 2: Unregularized cost.

$$J(\theta) = 19/(2 \times 5) = 19/10$$

Unregularized cost = 1.9

Step 3: Calculate $\sum \theta_j^2$.

$$2^2 + (-2)^2 + 1^2 = 4 + 4 + 1 = 9$$

Step 4: L2 penalty.

$$0.5/(2 \times 5) \times 9 = 0.5/10 \times 9 \\ = 0.45$$

Step 5: Regularized cost.

$$J_{\text{total}}(\theta) = 1.9 + 0.45$$

Regularized cost = 2.35

Step 6: Increase in cost = $2.35 - 1.9$

Increase due to regularization = 0.45

Q4. Ridge Cost Function

Given: $m = 10$, $SSE = 80$, $\theta_1 = 3$, $\theta_2 = -4$, $\theta_3 = 2$, $\lambda = 0.2$.

Step 1: Calculate the base cost.

$$SSE/(2m) = 80/(2 \times 10) \\ = 80/20$$

Base cost = 4

Step 2: Calculate $\sum \theta_j^2$.

$$3^2 + (-4)^2 + 2^2 = 9 + 16 + 4 \\ = 29$$

Step 3: Calculate Ridge penalty.

$$\lambda/(2m) \times 29 = 0.2/20 \times 29 \\ = 0.29$$

Step 4: Calculate Ridge cost.

$$J_{\text{Ridge}}(\theta) = 4 + 0.29$$

Final Answer: $J_{\text{Ridge}}(\theta) = 4.29$

Q5. Elastic Net Cost Function

Given: $m = 6$, $SSE = 48$, $\theta_1 = 2$, $\theta_2 = -3$, $\lambda = 0.6$, $\alpha = 0.7$.

Step 1: Calculate the base cost.

$$SSE/(2m) = 48/(2 \times 6) \\ = 48/12$$

Base cost = 4

Step 2: Calculate $\sum |\theta_j|$.

$$|2| + |-3| = 5$$

Step 3: Calculate $\sum \theta_j^2$.

$$2^2 + (-3)^2 = 4 + 9 = 13$$

Step 4: Calculate the Elastic Net bracket.

$$\begin{aligned} & \alpha \sum |\theta_j| + (1 - \alpha) \sum \theta_j^2 \\ &= 0.7(5) + 0.3(13) \\ &= 3.5 + 3.9 \\ &= 7.4 \end{aligned}$$

Step 5: Calculate the regularization term.

$$\begin{aligned} & \lambda/(2m) \times 7.4 = 0.6/12 \times 7.4 \\ &= 0.37 \end{aligned}$$

Step 6: Calculate total cost.

$$J_{EN}(\theta) = 4 + 0.37$$

Final Answer: $J_{EN}(\theta) = 4.37$